# 2.4 Solving Multi-Step Inequalities

Name _____ Period ____ Date _____

Algebra I 2.4 · Student sheet

**I can…**

- solve multi-step inequalities.
- identify inequalities with no solution or all real numbers as solutions.
- write and solve multi-step inequalities for real-life problems.

**Vocabulary** · fill in the missing word

**Multi-step inequality** — an inequality that _____ more than one step to solve

**All real numbers** — the _____ cancel and a true statement is left

**No solution** — the _____ cancel and a false statement is left

## Part 1 · Error hunt

Each solution has one mistake. Find it, explain it, and solve correctly.

| Student work | The mistake | Correct solution |
|---|---|---|
| $-2x + 3 > 11 \rightarrow -2x > 8 \rightarrow x > -4$ | | |
| $4(x - 1) \le 2x + 6 \rightarrow 4x - 1 \le 2x + 6 \rightarrow 2x \le 7 \rightarrow x \le 3.5$ | | |

## Part 2 · Notes: fill in the blanks

Reverse the symbol only when you multiply or divide by a _____.

Variables cancel, true statement: _____.

Variables cancel, false statement: _____.

## Part 3 · Examples (work them with the class)

**Example 1**

Solve $3x - 7 \ge 8$.

**Example 2**

Solve $5 - 2(x + 3) < 9$.

**Example 3**

Solve each inequality.

a. $4x + 2 > 4(x - 1)$ b. $3(x + 1) < 3x - 5$

**Example 4**

A gym charges $20 to join plus $15 per month. You have $110. For how many months can you be a member?

## Part 4 · Partner practice

**Partner 1**

Solve $-4x + 9 \leq 1$.

**Partner 2**

Solve $2(x - 3) \geq 2x - 6$.